Lecture 6

F-symbols and the pentagon

Objectives. *Lecture 5 gave each anyon model its particles and fusion rules, and a basis of fusion trees for the states of many particles. It said nothing about the associator, which Lecture 3 made part of the definition. In a fusion category the associator is a finite table of numbers, the F-symbols: the change of basis between two bracketings of three particles. Mac Lane's pentagon becomes a system of polynomial equations for them. This lecture writes it down and solves it for Fibonacci, where the fusion rules leave almost no freedom.*

By the end of this lecture you will be able to:

- Define the F-symbols on fusion trees and prove their unitarity for a unitary category with normalised vertices.
- Derive the pentagon equation from Mac Lane's pentagon.
- Compute how a change of vertices acts on the F-symbols, and find gauge invariants among them.
- Solve the pentagon for Fibonacci, and verify the two Ising solutions and why they are inequivalent.

6.1 F-moves

As in § 5.4, C is unitary and multiplicity-free. Its dagger gives positive inner products, and its associators are unitary. The vertices ψ_c^{ab} are normalised by (5.1). Three particles a, b, c with total charge d have two bases of fusion trees (Theorem 5.6, and its mirror image). The *left trees* fuse a and b first, through a channel e:

$$\begin{aligned} L_e &= (\psi_e^{ab} \otimes \mathrm{id}_c) \circ \psi_d^{ec} \\ &\in \mathrm{Hom}\big(d, (a \otimes b) \otimes c\big). \end{aligned}$$

The *right trees* fuse b and c first, through a channel f:

$$\begin{aligned} R_f &= (\mathrm{id}_a \otimes \psi_f^{bc}) \circ \psi_d^{af} \\ &\in \mathrm{Hom}\big(d, a \otimes (b \otimes c)\big). \end{aligned}$$

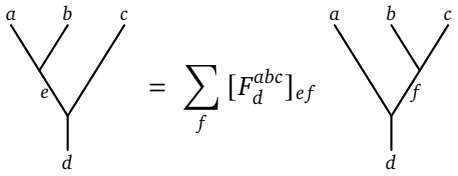
The associator $\alpha_{a,b,c}$ carries the first space to the second, and the R_f are a basis there.

Definition 6.1 (F-symbols)**.** The *F-symbols* $[F_d^{abc}]_{ef}$ are the numbers with

$$\alpha_{a,b,c} \circ L_e = \sum_f [F_d^{abc}]_{ef}\, R_f.$$

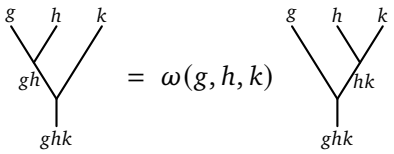
The *F-matrix* F_d^{abc} has rows indexed only by allowed left channels e and columns only by allowed right channels f, in the order of the simple objects. For sums over all simple labels, extend the indexed symbols by 0 when a tree does not exist. These zeros are not extra matrix rows or columns.

In pictures the associator is invisible (Lecture 3), and the definition is the *F-move*:



(6.1)

For Vec_G^ω every space is a line, and the associator of Lecture 3 gives a single F-symbol for each triple, $F^{ghk}_{ghk} = \omega(g,h,k)$:



(6.2)

For the twisted $\mathbb{Z}_2$ category of Example 3.9, this is $F_s^{sss} = -1$. An F-move is needed whenever an operation acts on two particles that the current tree does not fuse first, as an exchange will in Lecture 7.

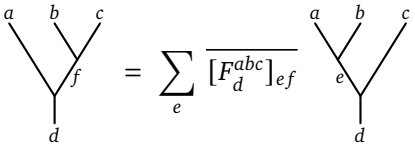
Proposition 6.2 (F is unitary). *Every F-matrix is unitary.*

Proof. By the bubble (5.1), applied twice,

$$\begin{aligned} L_e^\dagger L_{e'} &= \delta_{ee'} \sqrt{\frac{d_a d_b}{d_e}} \sqrt{\frac{d_e d_c}{d_d}}\ \mathrm{id}_d \\ &= \delta_{ee'} \sqrt{\frac{d_a d_b d_c}{d_d}}\ \mathrm{id}_d, \end{aligned}$$

and likewise $R_f^\dagger R_{f'} = \delta_{ff'} \sqrt{d_a d_b d_c / d_d}\ \mathrm{id}_d$. So both families are orthogonal with one common norm, and a unitary map between the two spaces has a unitary matrix in these bases. ∎

So the move back is the conjugate transpose, a sum over the left channel:

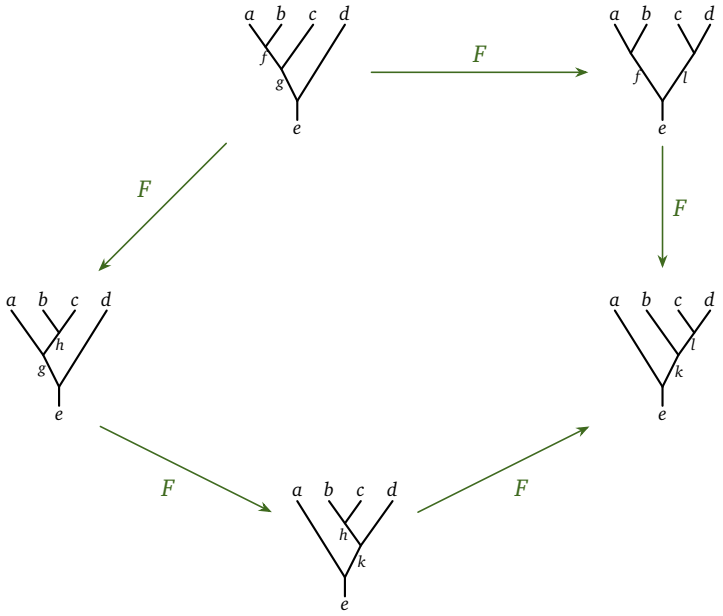


(6.3)

One more convention fixes the entries with a unit leg: $[F_d^{abc}]_{ef} = 1$ whenever a, b or c is 1 and the trees exist. Bonderson (p. 15) imposes it as the statement that fusing with the vacuum commutes with associativity, and notes that it can always be arranged by the choice of vertices.

6.2 The pentagon equation

Four particles a, b, c, d with total charge e have five bracketings, and the F-moves connect them. Labelling the inner edges as in the picture,



the two paths from the top-left tree to $a(b(cd))$ on the right must agree.

Theorem 6.3 (The pentagon equation). *The F-symbols of $\mathcal{C}$ satisfy, for all labels,*

$$[F_e^{fcd}]_{gl}\,[F_e^{abl}]_{fk} = \sum_h [F_g^{abc}]_{fh}\,[F_e^{ahd}]_{gk}\,[F_k^{bcd}]_{hl}.$$

Conversely · sketched, numbers $[F_d^{abc}]_{ef}$ for a fusion ring, invertible and equal to 1 on unit legs, that satisfy these equations define an associator which makes the skeletal category with these fusion rules monoidal.

Proof. Start from the tree $((ab)c)d$ with channels f for ab and g for $(ab)c$. The upper path first applies the F-move to f, c, d, which changes g into l, the channel of cd, with coefficient $[F_e^{fcd}]_{gl}$; then the F-move to a, b, l, which changes f into k, with coefficient $[F_e^{abl}]_{fk}$. This reaches the tree $a(b(cd))$ with channels l and k. The lower path applies F_g^{abc} inside, changing f into h, the channel of bc; then F_e^{ahd}, changing g into k; then F_k^{bcd}, changing h into l. It reaches the same tree after a sum over h. The two paths are the two sides of Mac Lane's pentagon (Definition 3.1), applied to the starting tree, so they are equal; the trees $a(b(cd))$ are a basis (Theorem 5.6, mirrored), so their coefficients are equal, which is the equation.

For the converse, define α on simple objects by the F-symbols and extend it by linearity. Mac Lane's pentagon then needs checking only on simple objects, where it is the equation, and the triangle holds by the unit legs. This is the argument of Theorem 3.8. ∎

For Vec_G^ω the equation at $(a,b,c,d) = (g,h,k,m)$ has $f = gh$, $l = km$ and one term on the right, and it reads

$$\omega(gh,k,m)\,\omega(g,h,km) = \omega(g,h,k)\,\omega(g,hk,m)\,\omega(h,k,m),$$

the cocycle condition of Theorem 3.8. In general it is a system of polynomial equations, one for each choice of the nine labels, in the unknown F-symbols; for Fibonacci it has twelve distinct non-trivial equations in five unknowns.

6.3 Gauge and Ocneanu

The vertices were chosen up to a phase (Lecture 5), and the F-symbols depend on the choice; any non-zero rescaling of the vertices changes them in the same way.

Theorem 6.4 (Gauge transformations). *Let $\psi_c^{ab} = u_c^{ab}\,\tilde{\psi}_c^{ab}$ with $u_c^{ab} \neq 0$ and $u_a^{1a} = u_a^{a1} = 1$. In the new vertices*

$$[\tilde{F}_d^{abc}]_{ef} = \frac{u_d^{af}\,u_f^{bc}}{u_e^{ab}\,u_d^{ec}}\,[F_d^{abc}]_{ef},$$

and the $\tilde{F}$ satisfy the pentagon equation and are 1 on unit legs. If every $|u_c^{ab}| = 1$, they are unitary.

Proof. Substitute into Definition 6.1: $L_e = u_e^{ab}\,u_d^{ec}\,\tilde{L}_e$ and $R_f = u_f^{bc}\,u_d^{af}\,\tilde{R}_f$. The pentagon is a statement about α, which has not changed, written in the new trees. If $|u| = 1$ the new vertices still satisfy (5.1), so Proposition 6.2 applies to them. With a unit leg the factor is 1. ∎

A number that does not change is a *gauge invariant*, and only those describe the category: two solutions related by a gauge transformation are the same category. An entry whose factor is 1 is invariant. For a self-dual a, the entry $[F_a^{aaa}]_{11}$ has the factor $u_a^{a1}\,u_1^{aa}/(u_1^{aa}\,u_a^{1a}) = 1$, and it is the Frobenius–Schur indicator of Lecture 4 divided by the dimension (Bonderson (2.23) · *cited*):

$$\kappa_a = d_a\,[F_a^{aaa}]_{11}. \tag{6.4}$$

For that twisted $\mathbb{Z}_2$ category, $\kappa_s = \omega(s,s,s) = -1$.

For Vec_G^ω the factor of Theorem 6.4 on F^{ghk} is $u(g,hk)\,u(h,k)/(u(g,h)\,u(gh,k))$, writing $u(g,h) = u_{gh}^{gh}$: the cocycle ω changes by a coboundary. With $j = u^{-1}$ this is exactly the change $\omega' = \omega\,(\delta j)^{-1}$ made by a monoidal functor that is the identity on objects (Definition 3.10). A gauge transformation is such a functor, and Proposition 3.11 found its invariant on $\mathbb{Z}_2$: $\omega(s,s,s)$, which is κ_s.

The pentagon solutions for fixed fusion rules are rigid.

Theorem 6.5 (Ocneanu rigidity · *cited*). *A fusion category admits no non-trivial deformations, and there are only finitely many fusion categories with a given fusion ring, up to equivalence (EGNO Thm. 9.1.4).*

In F-symbols: the pentagon for a given fusion ring has only finitely many solutions up to gauge transformations with $u \neq 0$ (Bonderson, p. 33).

6.4 Fibonacci solved

Fibonacci has one non-trivial fusion, $\tau \otimes \tau \cong 1 \oplus \tau$. With the unit legs fixed, the unknowns are the number $x = [F_1^{\tau\tau\tau}]_{\tau\tau}$ and the 2×2 matrix

$$F_\tau^{\tau\tau\tau} = \begin{pmatrix} A & B \\ C & D \end{pmatrix},$$

rows and columns in the order $1, \tau$. By Theorem 6.4, with $u_1^{\tau\tau} = p$ and $u_\tau^{\tau\tau} = q$, the entries x, A and D are invariant, while

$$\tilde{B} = \frac{q^2}{p}\,B, \qquad \tilde{C} = \frac{p}{q^2}\,C,$$

so BC is invariant and the ratio B/C is not. Write the pentagon equation with $a=b=c=d=\tau$. To read an instance, take $e=f=1$, $g=k=\tau$ and $l=1$. On the left, $F_1^{1\tau\tau}$ and $F_1^{\tau\tau 1}$ have a unit leg, so the left side is 1. On the right, $h=1$ gives $[F_\tau^{\tau\tau\tau}]_{11}[F_1^{\tau 1\tau}]_{\tau\tau}[F_\tau^{\tau\tau\tau}]_{11}=A^2$ and $h=\tau$ gives $B\,x\,C$: so $1=A^2+xBC$. The following instances list (e,f,g,k,l) beside each equation, in the order used below:

$$\begin{aligned}
(\tau,1,\tau,1,\tau):&\quad A=xBC,\\
(\tau,\tau,1,1,\tau):&\quad BC=x^2A,\\
(1,1,\tau,\tau,1):&\quad xBC+A^2=1,\\
(\tau,1,\tau,\tau,1):&\quad A^2+BCD=0,\\
(\tau,\tau,1,\tau,\tau):&\quad BD=xBD,\\
(1,\tau,\tau,\tau,\tau):&\quad x^2=xD^2+BC.
\end{aligned}$$

Theorem 6.6 (Fibonacci's F-symbols). *Every solution has $x=1$, $A\in\{\varphi^{-1},-\varphi\}$, $D=-A$ and $BC=A$. The unitary ones have $A=\varphi^{-1}$, and in the gauge $B=C$*

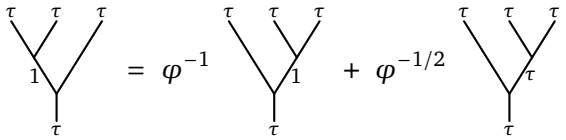
$$F_\tau^{\tau\tau\tau}=\begin{pmatrix}\varphi^{-1} & \varphi^{-1/2}\\ \varphi^{-1/2} & -\varphi^{-1}\end{pmatrix},$$

unique up to the gauge $B,C\mapsto -B,-C$.

Proof. The first two equations give $A=x^3A$. If $A=0$, then $BC=0$ and the third reads $0=1$; so $A\neq 0$, $x^3=1$, and B and C are non-zero. If $D=0$, the fourth gives $A^2=0$; so $D\neq 0$, and the fifth gives $x=1$. Now the first gives $BC=A$, the third $A^2+A=1$, and the fourth $A^2+AD=0$, so $D=-A$; the sixth then holds. The roots of $A^2+A=1$ are φ^{-1} and $-\varphi$, since $\varphi^2=\varphi+1$.

A unitary matrix has entries of modulus at most 1, which excludes $A=-\varphi$. For $A=\varphi^{-1}$, a unit first row needs $|B|^2=1-\varphi^{-2}=\varphi^{-1}$. The gauge with $p=1$ and $q^4=C/B$ makes $B=C$, and then $B^2=A$ gives $B=C=\pm\varphi^{-1/2}$; the sign is the gauge $q\mapsto iq$. The matrix is unitary, and every other pentagon equation holds (`scripts/check-l6-algebra.py` checks them all). ∎

In pictures, three τ's whose first two fuse to the vacuum are



(6.5)

The gauge-dependent entries are not physics: $B=\varphi^{-1/2}$ is a choice, while $x=1$, A, D and BC are forced. By (6.4), $A=\kappa_\tau/d_\tau$; the unitary root $A=\varphi^{-1}$, with $d_\tau=\varphi$ from Lecture 5, gives $\kappa_\tau=1$.

The rejected root is not an error. With $A=-\varphi$ the matrix, with $B=C=\sqrt{-\varphi}$, still solves every pentagon equation and squares to the identity. It is the *Yang–Lee* category, the Galois conjugate $\sqrt{5}\mapsto-\sqrt{5}$ of Fibonacci, with $d_\tau=1/A=-\varphi^{-1}$, the other root of $d^2=1+d$ that Lecture 5 set aside. It is a fusion category (EGNO Exer. 8.18.7 · *cited*), but not a unitary one, so it is no anyon model (Sheet 6). Up to gauge, then, the fusion rule $\tau\otimes\tau\cong 1\oplus\tau$ allows exactly these two categories, as Ocneanu's theorem promised: finitely many.

6.5 Ising: κH verified; κ gauge invariant

Ising has eleven F-matrices with no unit leg: $F_\sigma^{\sigma\sigma\sigma}$, which is 2×2 in the channels $1,\psi$, and ten numbers. Let H be the Hadamard matrix of *Quantum Computing* (Eq. (2.1)),

$$H=\frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix}.$$

Theorem 6.7 (Ising's F-symbols · *proved by verification*)**.** *For $\kappa=\pm1$, the F-symbols*

$$F_\sigma^{\sigma\sigma\sigma}=\kappa H,$$
$$F_\psi^{\sigma\psi\sigma}=F_\sigma^{\psi\sigma\psi}=-1,$$

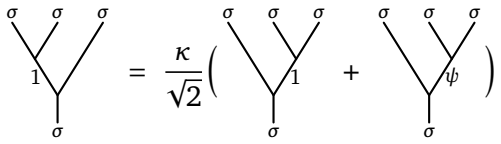
with every other one equal to 1, solve the pentagon equation. The two solutions are not related by any gauge transformation.

Proof. The pentagon equation is a finite list, and `scripts/check-l6-algebra.py` checks all 3^9 instances for both signs. One instance shows where the -1 comes from: take $a=b=c=d=\sigma$, $e=\psi$, $f=1$, $g=k=\sigma$ and $l=1$. The left side vanishes, since f and l would have to fuse to ψ, and $1\otimes 1\cong 1$. The right side has the two terms $h=1,\psi$:

$$0=\frac{\kappa^2}{2}\Big(1+[F_\psi^{\sigma\psi\sigma}]_{\sigma\sigma}\Big).$$

Since σ is self-dual, (6.4) gives the gauge invariant $d_\sigma[F_\sigma^{\sigma\sigma\sigma}]_{11}=\sqrt{2}\,\kappa/\sqrt{2}=\kappa$, which differs between the two solutions. ∎

In pictures, three σ's whose first two fuse to the vacuum are



(6.6)

A monoidal equivalence between the two would fix σ, the only simple object of dimension $\sqrt{2}$, and ψ. Identifying each image with the object, it carries each vertex to a multiple of a vertex, and its compatibility with the associators (Definition 3.10) becomes the formula of Theorem 6.4 with some $u\neq 0$ · *sketched*. That leaves κ alone.

So the Ising fusion rules carry two inequivalent fusion categories, told apart by the Frobenius–Schur indicator of σ. That there are no others is a theorem of Tambara and Yamagami · *cited* (EGNO Prop. 9.14.9). The course's *Ising* is $\kappa=+1$; the solution $\kappa=-1$ is the fusion category of $\mathrm{SU}(2)_2$, whose σ is spin-$\frac{1}{2}$ with $\kappa=-1$, as in Lecture 4. The two have the same fusion ring, as Lecture 5 found, and differ in their F-symbols (Simon, p. 98). As gates, H and $-H$ agree up to a phase; as F-symbols they give inequivalent categories. The printed entries are in this gauge: unlike κ, the off-diagonal entries of κH change under $u_\psi^{\sigma\sigma}\mapsto iu_\psi^{\sigma\sigma}$, and $F_\psi^{\sigma\psi\sigma}$, with the (ψ,ψ) entry of κH, under $u_\sigma^{\psi\sigma}\mapsto -u_\sigma^{\psi\sigma}$.

Summary.

- *Handed on:* the F-symbols are the associator on fusion trees, $\alpha\circ L_e=\sum_f[F_d^{abc}]_{ef}R_f$. For a unitary category with a norm-preserving vertex gauge, the admissible F-matrices are unitary. Entries on allowed unit legs are 1 (Definition 6.1 and Proposition 6.2).

- *The pentagon:* Mac Lane's pentagon is a polynomial system in the F-symbols (Theorem 6.3); for Vec_G^{ω} it is the cocycle condition.
- *Gauge:* changing vertices rescales F (Theorem 6.4); only invariants, such as $\kappa_a = d_a[F_a^{aaa}]_{11}$, are physics. Ocneanu: finitely many solutions up to gauge (Theorem 6.5).
- *Fibonacci:* $F_\tau^{\tau\tau\tau} = \left(\begin{smallmatrix} \varphi^{-1} & \varphi^{-1/2} \\ \varphi^{-1/2} & -\varphi^{-1} \end{smallmatrix}\right)$, unique up to gauge among unitary solutions; Yang–Lee is its non-unitary conjugate (Theorem 6.6).
- *Ising:* $F_\sigma^{\sigma\sigma\sigma} = \kappa H$, two inequivalent solutions; the course's Ising has $\kappa = +1$ (Theorem 6.7).
- *Next:* Lecture 7 adds the braiding: R-symbols, the hexagon equations and the topological spin.

References

Kitaev, *Anyons in an exactly solved model and beyond*, cond-mat/0506438: the F-move, eq. (177), p. 68; the pentagon, eq. (179), p. 69; Ocneanu rigidity, p. 89, and gauge transformations, eqs. (239)–(242), p. 90; the Ising F-matrix κH, eq. (86), p. 38. Bonderson, PhD thesis (Caltech, 2007): F-moves and their unitarity, (2.6)–(2.7), pp. 14–15; unit legs, p. 15; κ_a and F, (2.23), p. 20; the multiplicity-free pentagon and gauge, (2.77) and (2.80)–(2.81), p. 33; Fibonacci and Ising data, pp. 82–84. Simon, *Topological Quantum*: F-matrices, (9.1), p. 113; Fibonacci, (9.3), p. 115; the pentagon and Ocneanu, (9.7), p. 117; gauge, (9.8), p. 118; Ising and $\mathrm{SU}(2)_2$, p. 98, footnote 7. Etingof, Gelaki, Nikshych and Ostrik, *Tensor Categories*: Ocneanu rigidity, Thm. 9.1.4, p. 276; Ising categories, Prop. 9.14.9, p. 315; the Yang–Lee categories, Exer. 8.18.7, p. 236, and Exer. 9.4.6(iii), p. 283.